

HR DATA ANALYTICS & ATTRITION FRAMEWORK

Data Warehouse Architecture & Interactive Visualization Project Documentation

Author / Data Analyst: Project Portfolio Repository Add-On

Tools Utilized: PostgreSQL (Data Aggregations, Matrix Pivots) | Tableau Desktop Production Engine

Data Scope: 1,470 Active Corporate Records | Multi-Department Integration Pipeline

Classification: Professional Portfolio Presentation & Deployment Asset

1. Executive Summary & Core KPIs

This project establishes a high-performance relational analytics architecture designed to process complex human resource data, isolate hidden drivers behind organizational attrition, and engineer an intuitive platform for business intelligence tracking. Using an advanced tech stack combining **PostgreSQL data engineering** and **Tableau visualization modeling**, raw data files were processed into validated high-impact metrics.

Metric Parameter	Empirical Value	Strategic Significance
Total Employee Headcount	1,470 Records	Baseline operational organizational capacity.
Total Attrition Count	237 Departures	Historical subset of realized workforce attrition.
Global Attrition Rate	16.12%	Core organization benchmark attrition factor.
Active Workforce Strength	1,233 Employees	Current cross-functional staff count.
Mean Employee Age	37 Years Old	Demographic benchmark for target programs.

2. Deep-Dive Metrics & Strategic Insights

2.1 Departmental Turnover Distribution

- **Research & Development (R&D):** Holds the highest absolute attrition burden across the organization at 133 absolute departures, constituting 56.12% of the aggregate churn metric.
- **Sales Operations:** Exposes a highly critical structural attrition acceleration factor of 20.63%, accumulating 92 absolute departures.
- **Human Resources:** Accounts for a marginal load of 12 internal employee turnover events (5.06% of macro totals).

2.2 Demographic Churn Intersections

- **The Age-Band Exposure Window:** Employee turnover heavily clusters inside the 25–34 age segment, yielding an elevated localized attrition factor of 29.11% (112 distinct employees).
- **Gender Segment Allocations:** Out of 237 overall attrition events, Male staff events represent 150 instances while Female staff instances compose 87 entries.
- **High School Cohort Deep-Dive:** Isolating high-school level workers within the data architecture reveals 20 Male departures and 11 Female departures.

2.3 Education & Satisfaction Matrix Findings

- **Education Churn Alignments:** Employees with Life Sciences backgrounds show the highest raw attrition count (89), followed closely by Medical cohorts (63).

- **Satisfaction Disparities:** Cross-tabulation matrices map localized retention variations across discrete operational roles, pointing to acute friction within frontline client roles.

3. Technical SQL Data Engineering Pipeline

To guarantee complete analytical data integrity between core tables and transactional visual dashboards, a modular suite of production-grade SQL code blocks was deployed. The schema handles everything from conditional aggregations to complex matrix pivots via the PostgreSQL tablefunc runtime environment.

3.1 Dynamic Macro Attrition Rate Engineering

```
-- Computes global attrition rates against shifting operational filters
SELECT ROUND(((SELECT COUNT(attrition) FROM hrdata WHERE attrition = 'Yes') /
              SUM(employee_count)) * 100, 2) AS validated_attrition_rate
FROM hrdata;
```

3.2 Gender Segment Isolation Matrix Query

```
-- Aggregates turnover records for educational sub-cohorts
SELECT gender, COUNT(attrition) AS conditional_attrition_count
FROM hrdata
WHERE attrition = 'Yes' AND education = 'High School'
GROUP BY gender
ORDER BY COUNT(attrition) DESC;
```

3.3 Production-Grade Job Satisfaction Cross-Tabulation

```
-- Resolves data-shifting issues by injecting explicit 2-argument categories
CREATE EXTENSION IF NOT EXISTS tablefunc;

SELECT *
FROM crosstab(
  'SELECT job_role, job_satisfaction, SUM(employee_count)::numeric
   FROM hrdata
   GROUP BY job_role, job_satisfaction
   ORDER BY job_role, job_satisfaction',
  'VALUES (1), (2), (3), (4)'
) AS ct(job_role VARCHAR(50), level_1 numeric, level_2 numeric, level_3 numeric, level_4 numeric)
ORDER BY job_role;
```

4. Deployment & GitHub Version Control Workflow

For professional distribution, this analytics framework is structured into a clean, reproducible version-controlled repository architecture. Below is the production workflow utilized to deploy and maintain codebase files:

```
# Initialize project structure tracking
git init

# Track data assets, analytical workbooks, and schema scripts
git add .

# Log initial system deployment commit
git commit -m "Initial commit: Add validated SQL workflows and dashboard workbooks"

# Configure main branch deployment path
git branch -M main
git remote add origin https://github.com/yourusername/hr-analytics-dashboard.git
git push -u origin main
```